IEEE Open Journal of the Computer Society

Received 18 March 2025; revised 11 June 2025; accepted 13 June 2025. Date of publication 25 June 2025; date of current version 25 July 2025. The review of this article was arranged by Associate Editor Peng Li.

Digital Object Identifier 10.1109/OJCS.2025.3582726

Enhancing Pneumonia Diagnosis Through AI Interpretability: Comparative Analysis of Pixel-Level Interpretability and Grad-CAM on X-ray Imaging With VGG19

MOHAMMAD ENNAB AND HAMID MCHEICK (Fellow, IEEE)

Department of Computer Sciences and Mathematics, University of Québec at Chicoutimi, 555 Bd de University, Chicoutimi, QC G7H 2B1, Canada

CORRESPONDING AUTHOR: MOHAMMAD ENNAB (email: mennab@etu.uqac.ca).

ABSTRACT Pneumonia is a leading cause of morbidity and mortality worldwide, necessitating timely and precise diagnosis for effective treatment. Chest X-rays are the primary diagnostic tool, but their interpretation demands substantial expertise. Recent advancements in AI have shown promise in enhancing pneumonia detection from X-ray images, yet the opacity of deep learning models raises concerns about their clinical adoption. Interpretability in AI models is vital for fostering trust among healthcare professionals by providing transparency in decision-making processes. This study conducts a comparative analysis of two interpretability methods, Pixel Level Interpretability (PLI) and Gradient-weighted Class Activation Mapping (Grad-CAM), in the context of pneumonia classification using VGG19 on X-ray datasets. The research includes an experiment involving three distinct X-ray datasets. VGG19 is applied to classify a query image, and both PLI and Grad-CAM are used to interpret the classification decisions. The study evaluates these interpretability methods across multiple dimensions: computational efficiency, diagnostic performance, explanation continuity, calibration accuracy, robustness to training parameters, and feedback from medical experts. Our findings aim to determine which interpretability technique offers a more clinically meaningful explanation, balancing computational feasibility and diagnostic reliability. This study contributes to the development of explainable AI in healthcare, supporting the integration of trustworthy AI systems in clinical environments for enhanced pneumonia diagnosis.

INDEX TERMS AI interpretability, deep learning, grad-CAM, machine learning, medical imaging, PLI, pneumonia, VGG19.

I. INTRODUCTION

Chest X-ray diagnosis remains the first line of diagnosis for pneumonia, yet operation requires great expertise [1]. Recently, AI-driven models have been performing reasonably in classifying pneumonia from chest X-rays, hence supporting quicker or potentially more accurate diagnoses [2]. However, healthcare requires better interpretability, where AI decisions need to be transparent and understandable for the clinicians if they are to trust and eventually rely on such models [3]. Interpretability in AI models provides insight into the decision-making process of the model, enabling medical professionals to validate and understand a prediction for eventual facilitation of safety in patient care [4]. In the context of AI-assisted medical imaging, interpretability tools like PLI and Grad-CAM have become instrumental [5]. These interpretability models provide heatmaps, which serve as visual explanations by highlighting regions of the image that contribute to the classification outcome. Although PLI provides a fine-grained explanation with pixel-level relevance scores across an image [6], Grad-CAM creates high level heatmaps that indicate the important feature areas associated with specific classification categories [7]. Despite their utility, these two methods vary considerably in their granularity, computational requirements, and clinical utility. Our work compares PLI and Grad-CAM, for the interpretability of the of X-rays as either normal or infected with pneumonia using three different

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

datasets of normal and pneumonia-infected lung images. Our study is considered unique due to the stringent assessment of clinical applicability and efficiency that the two approaches bring to the enhancement of pneumonia diagnosis from X-ray imaging. This will be accomplished by critically evaluating each method's usefulness in terms of the following:

1) *Time Complexity Analysis*: Since healthcare settings are very time-sensitive and diagnosis needs to be done as fast as possible, the time complexity will play an important role in comparing the various methods of interpretability [8]. In this section, both PLI and Grad-CAM are compared for computational efficiency in terms of time taken and resources needed. It is necessary to estimate the time complexity since this has implications for the suitability of a particular method for real-time clinical applications [9].
2) *Performance Metrics Comparison*: To comprehensively evaluate each method, we will compare key performance metrics such as accuracy, precision, recall, and explanation continuity. This analysis provides a quantitative foundation for evaluating the diagnostic relevance and reliability of each interpretability method.
3) *Calibration Curves*: Calibration curves will be provided in order to obtain the calibration between the predicted probability and the actual outcome [10]. Well-calibrated models provide more reliable probability estimates, reducing the risk of over- or under-confidence in predictions [11]. In this section, the calibration of PLI and Grad-CAM is analyzed in order to determine which one of these two methods provides a more trustworthy interpretation of the actual diagnostic outcome.
4) *Feedback from Radiologists and Physicians*: This will be very crucial in understanding, from the practical point of view, how useful the interpretability methods are in the clinical setting. Qualitative insights will be gathered from healthcare professionals concerning clarity, intuitiveness, and diagnostic relevance of the outputs from PLI and Grad-CAM [12]. Such expert input would be an important component in determining if these methods align with or support the correct diagnosis upon comparison with expert interpretation, hence giving credence to the clinical validity of each interpretability method.

II. LITERATURE REVIEW

Recent advancements in AI for pneumonia diagnosis leverage deep learning architectures, primarily applied to chest X-rays and CT scans. Over time, there has been a growing demand for more interpretable deep learning models in medical imaging [13]. AI interpretability in healthcare has shown the highest prevalence, followed closely by heatmap-based models [13], [14]. Clinicians have urgent needs to make AI diagnosis both more transparent and understandable [4]. LIME, SHAP, Anchors, GRAD-CAM, PLI, and MAPLE are key models for interpreting AI predictions [13]. The interpretation methods involved in the study are PLI and GRAD-CAM. The article will review these interpretations concerning their effect on pneumonia classification using VGG19. It covers the theoretical underpinnings of both techniques with regard to interpretability, an overview of medical imaging applications, and previous comparative studies pointing out strengths and limitations of each method. This will finally form the foundation for a comparative analysis of PLI and Grad-CAM in terms of interpretability provided for VGG19-based X-ray image classification.

A. PNEUMONIA DIAGNOSIS

CNNs remain the most widely used architecture for pneumonia diagnosis due to their ability to automatically extract spatial hierarchies of features from CXR images [15]. Models like DenseNet-121, ResNet-50, and EfficientNet-B7 dominate benchmarks, achieving >90% accuracy in detecting pneumonia on datasets like NIH ChestX-ray14 and RSNA Pneumonia Challenge. EfficientNet balances accuracy and efficiency, making it suitable for deployment [16]. More recently, vision transformers (ViTs) have emerged as a powerful alternative to CNNs [17]. ViTs, such as the Transformer and Hybrid ViT-CNN models, leverage self-attention mechanisms to capture long-range dependencies in images, enhancing feature representation and diagnostic accuracy of pneumonia [18].

B. PIXEL LEVEL INTERPRETABILITY (PLI)

The PLI model, introduced by [6], pinpoints diagnostically significant regions in medical images through dynamic heatmaps that directly map pixel-level contributions to diagnostic conclusions. By isolating and visualizing high-resolution spatial patterns critical to decision-making, PLI delivers clinically actionable insights, enabling radiologists to validate AI-driven predictions against anatomically precise features. This fine-grained interpretability is particularly valuable for detecting subtle anomalies that could otherwise go unnoticed, such as early signs of pneumonia. The proposed heatmaps give pixel level interpretability and mark the region that has the highest influence on the classification outcomes, taking values over the range from 0.1 to 1 [5], allowing clinicians to correlate model predictions directly with precise anatomical features, thereby enhancing trust and facilitating AI integration into clinical workflows [19].

C. GRADIENT-WEIGHTED CLASS ACTIVATION MAPPING (GRAD-CAM)

Grad-CAM, proposed by [7], is another popular interpretability method that uses the gradients of target concepts flowing into the last convolutional layer to produce a localization map of the important regions in an image. Grad-CAM generates broader heatmaps, offering a high-level overview of the regions that influence AI decisions [5]. This makes it suitable for quickly identifying general patterns associated with diseases like pneumonia, especially in high-throughput diagnostic environments [20]. Grad-CAM is particularly effective in CNN architectures due to its ability to provide class-discriminative

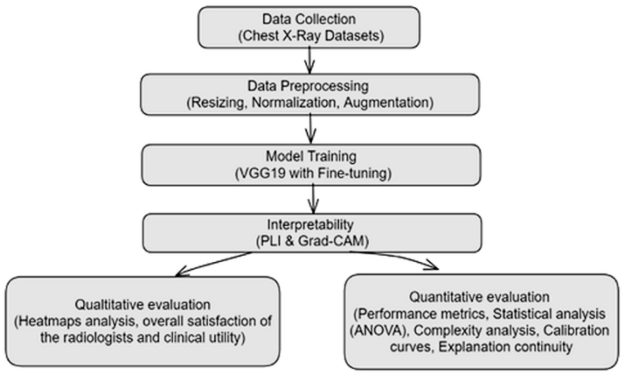


FIGURE 1. Workflow of the proposed model for pneumonia diagnosis and comparative analysis of PLI vs. Grad-CAM.

localization without requiring any changes to the model. In the context of medical imaging, Grad-CAM has been widely employed to interpret CNN-based predictions for conditions identified in X-rays and other imaging modalities, such as detecting pneumonia or identifying fractures [2]. Its flexibility and relatively low computational cost have made Grad-CAM a popular choice in various medical imaging studies [5].

D. VGG19 FOR MEDICAL IMAGING CLASSIFICATION

VGG19 is a CNN architecture that is known for its depth and simplicity, hence widely adopted in medical image classification tasks, given its proven performance and ease of implementation. Due to its layered structure, VGG19 can capture the features of complex images, which may turn it out to be suitable for medical datasets when nuanced visual features may specify disease conditions [21]. However, this also makes the model more complex and challenging to interpret, which raises the importance of applying PLI and Grad-CAM to understand its decision-making processes. Furthermore, ResNet introduces residual connections, which help in extracting more complex features, but this also makes feature attribution slightly more challenging [22]. SHAP values and feature importance mapping are often used to interpret ResNet-based models [23]. Unlike VGG19 and ResNet, YOLO is primarily used for pneumonia localization rather than classification [24]. It generates bounding boxes around affected regions, which provides a more intuitive visual interpretation but lacks pixel-level feature attribution methods like Grad-CAM.

III. EXPERIMENT DESIGN AND METHODOLOGY

This section describes various experiments related to the interpretability of the classification of chest X-ray images as either pneumonia positive or healthy using PLI and Grad-CAM. Fig. 1 illustrates an overview of the work: from data collection, preprocessing, and training a VGG19 model with fine-tuning, the application of interpretability models-PLI and Grad-CAM-and qualitative and quantitative evaluations. The complete schema ensures accurate diagnosis and the assessment of interpretability.

A. DATA COLLECTION AND PREPROCESSING

1) DATASETS

Three publicly available datasets of chest radiographs, comprising over 1000 labeled images (normal or pneumonia positive) were used, verified by radiologists.

The data described in this note can be accessed on Harvard Dataverse here [25]. The implementation of the PLI, VGG-19, and Grad-CAM models for pneumonia classification and interpretation is available at GitHub repository. While the datasets used are publicly available and radiologist-verified, they may not fully represent the diversity encountered in real-world clinical settings. For instance, these datasets may lack sufficient examples from underrepresented demographics or atypical presentations of pneumonia. Future studies will aim to validate the proposed methods on larger, more heterogeneous datasets to ensure robust generalizability across diverse patient populations.

2) PREPROCESSING STEPS

1) Resizing: The images were resized to 224x224 pixels to align with VGG19 input dimensions.
2) Normalization: Pixel intense
3) Data Augmentation: Techniques such as rotation, flipping, and brightness adjustment were applied to increase data variability and enhance robustness.
4) Noise Reduction: Gaussian filtering was applied to reduce image noise, focusing on diagnostically relevant details.

3) EXPERIMENTAL SETUP

1) Computing Environment: The experiments were conducted on NVIDIA Tesla V100 GPUs with TensorFlow 2.x as the primary deep learning framework, and data processing was automated through Python scripts.
2) Statistical Analysis: Performance Comparison: Paired t-tests were used to compare accuracy, precision, and recall metrics between PLI and Grad-CAM.
3) Confidence Intervals: Computed for each metric to ensure robustness and statistical validity.

B. WORKFLOW OF THE PLI ALGORITHM

Fig. 2 shows the flowchart of the PLI algorithm, the basis of the PLI model [6]. In essence, it describes the three basic steps of fuzzy logic: fuzzification, fuzzy inference, and defuzzification. This algorithm classifies images into one of three categories using 19 trapezoidal membership functions. These are optimally designed for the three potentially conflicting objectives of accuracy, interpretability, and generalizability. PLI proceeds systematically through the steps below:

1) Input Stage: Loads each X-RAY in grayscale or RGB format, establishing the base data structure.
2) Normalization: Adjusts pixel values to a consistent scale [0,1], minimizing variation due to differing imaging conditions.

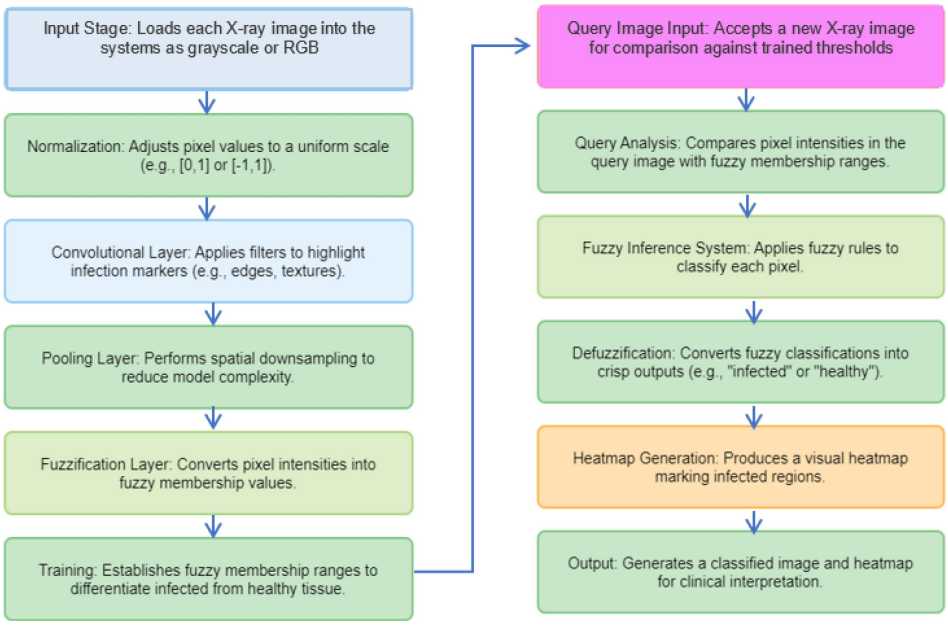


FIGURE 2. Workflow of the PLI model.

3) Convolutional Layer: Applies kernels or filters to highlight edges, textures, and other features correlated with infection markers.
4) Pooling Layer: Conducts spatial down sampling using max or average pooling, retaining significant features while reducing computational complexity.
5) Fuzzification Layer: Converts pixel intensities to fuzzy membership values, assigning degrees like high or low infection likelihood for interpretability.
6) Training: Establishes fuzzy membership ranges based on pixel-level data, defining fuzzy thresholds to distinguish infected from healthy regions.
7) Query Image Input: Introduces a new X-RAY, preprocessed and analyzed based on trained fuzzy thresholds.
8) Query Analysis: Compares pixel intensities with trained fuzzy membership ranges to evaluate infection likelihood.
9) Fuzzy Inference System: Applies fuzzy rules, such as if high infection likelihood, classify as infected, assigning classifications per pixel.
10) Defuzzification: Converts fuzzy classifications into definitive outputs, like infected or healthy.
11) Heatmap Generation: Creates a visual representation, highlighting high-risk or infected regions through color gradients for clinical interpretation.
12) Output: Generates the heatmap for clinical use.

C. VGG19 ARCHITECTURE AND TRAINING SETUP

Fig. 3 outlines the complete workflow for classifying pneumonia infection in the query X-ray image using VGG19 [21]. It begins with data preparation, where the dataset is composed and preprocessed through resizing, normalization, and data augmentation. In the model setup phase, the VGG19 model, pre-trained on ImageNet, is configured using transfer learning with specific training parameters. The experiment design involves data splitting into training, validation, and testing sets, followed by training and evaluation of performance metrics such as accuracy, precision, recall, and F1 score. VGG19 follows a series of well-defined steps:

1) The pretrained VGG19 model, initialized with ImageNet weights, was used as the base model for the classification task. Additional fully connected layers were added to adapt the model for binary classification (PNEUMONIA-positive vs. normal).
2) Data Splitting: The dataset was divided into training (70%), validation (15%), and test (15%) sets.
3) Transfer Learning: The convolutional layers of the VGG19 model were initially frozen to retain general feature extraction capabilities. Fine-tuning of the higher layers was conducted based on performance, allowing the model to specialize in the binary classification task.
4) Batch Size: The model was tested with batch sizes {32, 64, 128}, and a batch size of 64 was selected as it provided the best trade-off between memory efficiency and gradient stability, resulting in faster convergence and improved performance metrics.
5) Learning Rate: After extensive tuning between learning rates {1e-2, 1e-3, 1e-4}, a learning rate of 1e-4 was chosen. This rate struck the optimal balance between convergence speed and stability, preventing oscillations and divergence during training.
6) Number of Epochs: The model was trained for a maximum of 50 epochs. Early stopping was implemented with a patience of 10 epochs, meaning training

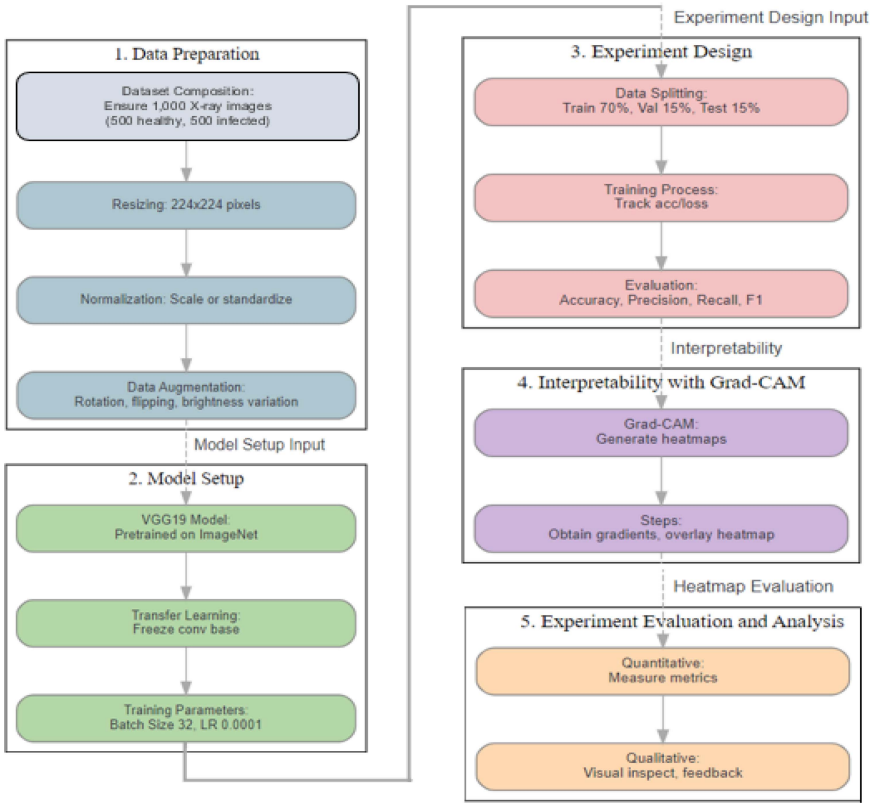


FIGURE 3. Workflow of Grad-CAM and VGG-19.

would halt if no improvement was observed for 10 consecutive epochs in the validation set. Optimal model weights were obtained at epoch 30, when the validation loss plateaued.

7) Dropout Rate: Dropout regularization was applied to prevent overfitting. Various dropout rates (0.2, 0.3, 0.4) were tested, with a dropout rate of 0.3 proving to be the most effective for maintaining high validation accuracy and generalization.
8) Weight Initialization: He initialization was used for the weights, particularly in layers with ReLU activations. This initialization method ensured that the variance of the outputs remained stable across layers, improving convergence speed and overall learning efficiency.
9) Optimizer: The Adam optimizer, with default parameters ($\beta 1 = 0.9$, $\beta 2 = 0.999$), was employed for its adaptive learning rate capabilities. This optimizer handles sparse gradients effectively, contributing to more stable convergence.
10) Loss Function: Binary cross-entropy was selected as the loss function due to the binary classification nature of the task (PNEUMONIA-positive vs. normal).
11) Choice facilitated optimization for probabilistic outputs, enabling the model to differentiate between the two distinct classes.
12) L2 Regularization: To further mitigate overfitting, L2 regularization was applied with a λ value of 0.001. This parameter was determined through cross-validation, and L2 regularization proved effective in reducing model complexity and improving generalizability.
13) Training Process: The model was trained with early stopping to prevent overfitting, optimizing both accuracy and generalizability.
14) Generate Predictions: The X-RAY passed through VGG19 and obtained the predicted classifications.

D. WORKFLOW OF THE GRAD-CAM

Fig. 3 shows the steps for utilizing the Grad-CAM to enhance interpretability by generating heatmaps that visually highlight regions influencing the model's decisions. The overall methodology of Grad-CAM involves a systematic progression through these stages [7]:

1) Compute Gradients: Gradients of the predicted class with respect to the last convolutional layer are used to create a Grad-CAM heatmap.
2) Overlay Heatmap: The heatmap is overlaid onto the original X-RAY, highlighting regions influential to the model's classification.

IV. EXPERIMENT EVALUATION AND ANALYSIS

A. QUALITATIVE ANALYSIS

1) PLI HEATMAP

Fig. 4 illustrates the heatmap output generated by PLI model. The red-highlighted region represents the areas of highest activation intensity, with a peak intensity of 88% corresponding

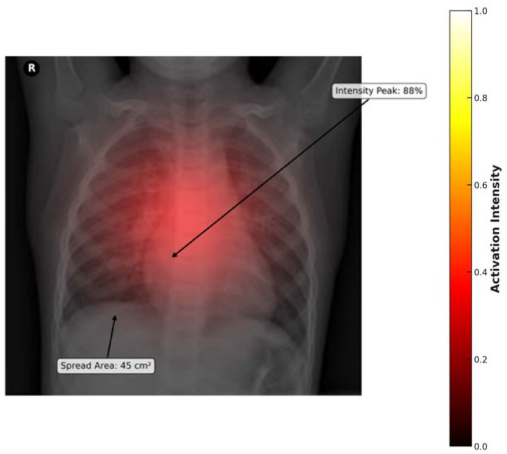


FIGURE 4. PLI heatmap with calculated metrics.

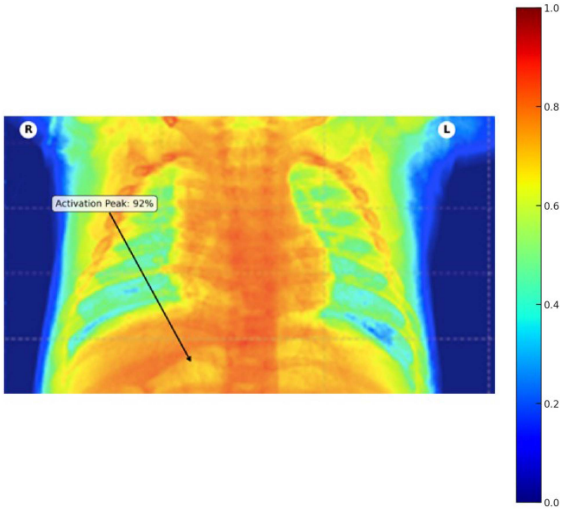


FIGURE 5. Grad-CAM heatmap with calculated metrics.

to the suspected infected tissue. The spread area, approximately 45 cm^2, delineates the extent of activation, while the accompanying color bar provides a scale for activation intensity, ranging from 0.0 (low) to 1.0 (high). This visualization allows for detailed insight into the patterns and correlations that are crucial for identifying potential malignancies. The model represents fuzzy membership values as heatmaps, where lighter colors denote higher membership values (indicating greater weight and likelihood of malignancy) and darker colors denote lower values (minimal risk). This is an intuitive but informative output. This approach enhances interpretability in classification decisions by clearly depicting complex, ambiguous relationships in a format easily accessible to the viewer [6].

2) GRAD-CAM HEATMAP

Fig. 5 displays the activation regions in the chest X-ray as identified by Grad-CAM, with an activation peak of 92% highlighted. The color-coded map represents the activation intensity, with regions in red indicating higher activation; the red areas denote the region of highest importance and relative weights which contributed most toward the result of classification and may indicate potential markers of infection or abnormality. This gradient of color-from red for high importance to blue for low importance-provides a visualization of this range of importance determined by the model and, further, provides an understandable visualization of how deep learning architectures, such as VGG19, classify pneumonia cases. Such a visualization-in highlighting the points of the model's focus-provides insight not only into what the model has determined but why it has done so, an integral part of the interpretation process for the clinicians in understanding AI-driven diagnostic assessments.

3) FINDINGS OF EXPERT EVALUATION FROM CLINICAL VALIDATION

The clinical validation of PLI and Grad-CAM involved systematic feedback from radiologists and physicians, who evaluated the tools on accuracy, interpretability, localization precision, workflow integration, and clinical utility. Seven clinicians participated in a clinical evaluation session, including five board-certified radiologists and two senior respiratory medicine specialists. The radiologists had a mean professional experience of 11.4 years (between 7 and 18 years) with all having ongoing clinical duties within diagnostic imaging. Two were thoracic imaging subspecialists. The physicians each had over 12 years of experience in the diagnosis and treatment of pulmonary disease. Collectively, their opinions represented a multidisciplinary perspective regarding the clinical interpretability, diagnostic intuitiveness, and real-world usefulness of the PLI and Grad-CAM models. Radiologists highlighted PLI's strength in pinpointing subtle, localized pathologies, particularly in early-stage cases, while Grad-CAM was praised for providing a quick, intuitive overview useful for screening large volumes of images. Table 1 summarizes expert satisfaction, showing high ratings for PLI's accuracy and precision in localization compared to Grad-CAM's broader focus. Both models enhanced interpretability and trust in AI-assisted decision-making, though challenges remain in integrating these tools seamlessly into clinical workflows, underscoring the need for further development.

a) Detailed feedback distribution: Fig. 6 illustrates the distribution of satisfaction levels among of radiologists and physicians. Accordingly, most experts stated high satisfaction levels because of the accuracy and fine localization precision of PLI. In contrast, the satisfaction about GRAD-CAM is equally distributed; the higher share belongs to moderate satisfaction because this method is based on coarser localization.

b) Recommendations based on feedback: Based on the analysis of radiologists' and physicians' feedback, several recommendations were made to improve the clinical utility of PLI and GRAD-CAM:

1) Enhanced Localization for GRAD-CAM: Increasing the focus on improving GRAD-CAM 's ability to detect

TABLE 1. Expert Evaluation of PLI and GRAD-CAM Based on Key Clinical Criteria

Evaluation Criteria	PLI Satisfaction	GRAD-CAM Satisfaction	Comments and Analysis
Accuracy and diagnostic reliability	Highly satisfied	Moderately satisfied	Radiologists showed a high degree of satisfaction with the accuracy of PLI in determining pathologically relevant areas, in particular for imaging modalities such as mammograms and X-RAYs. PLI was able to give more importance to some patches, hence the possibility of detecting small lesions with much higher resolution. GRAD-CAM had been helpful to indicate larger areas of interest but has some issues in pinpointing small and rather subtle anomalies, therefore achieving a moderate level of satisfaction
Interpretability and clarity	Highly satisfied	Highly satisfied	Both PLI and GRAD-CAM were praised for their ability to enhance interpretability. PLI's pixels, based approach offered clear, localized insights that radiologists found intuitive to review. GRAD-CAM 's heatmaps provided a clear visual understanding of the AI's decision, making process, allowing radiologists to trust the model's reasoning. The clarity provided by these methods significantly improved the experts' confidence in the AI tools
Localization precision	Highly satisfied	Moderately satisfied	PLI excelled in localizing small, precise areas of interest, this is very helpful in the case of early disease detection. According to the radiologists, PLI was able to direct them to regions that might get missed while interpreting a whole image. Sometimes, GRAD-CAM failed to give the required sharpness on the detection of pathologies of smaller dimensions. Though it was performing quite well on broad areas, it was not able to precisely localize fine details in certain cases, specially when the presentation of a case was complex.
Workflow integration	Moderately satisfied	Moderately satisfied	Both PLI and GRAD-CAM were seen as valuable tools, but their integration into existing workflows presented some challenges. Radiologists indicated that the time spent reviewing PLI model or heatmaps GRAD-CAM could extend the diagnostic process. To achieve full workflow efficiency, further refinements in interface design and output interpretation speed were necessary. However, both tools were appreciated as potential aids in second, opinion scenarios.
Overall satisfaction and clinical utility	Highly satisfied	Satisfied	Overall, PLI received the highest praise for its clinical utility, with radiologists noting it provided a reliable way to identify and confirm suspicious areas in imaging. GRAD-CAM received positive responses but at a more moderate level of satisfaction, as it did provide insight into how the model came up with its conclusion, though its larger focus areas meant that further refinement was necessary before high, stakes diagnoses could be fully placed in its trust.

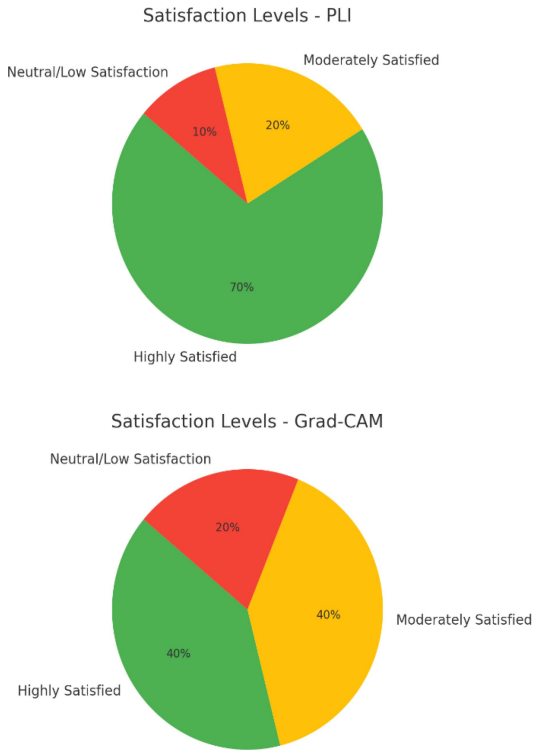


FIGURE 6. Comparison of user satisfaction levels for PLI and Grad-CAM interpretability models.

smaller, more subtle features would increase its reliability in clinical settings.

2) Improved Workflow Integration: Streamlining the process of reviewing heatmaps, perhaps through better interface design or faster data processing, would make PLI and GRAD-CAM more practical for everyday use.
3) Combination of Methods: Radiologists suggested that combining PLI's granular detail with GRAD-CAM 's broad region analysis could offer a more robust, complementary approach, leading to a higher degree of diagnostic confidence.

Overall, the clinical validation of PLI and GRAD-CAM revealed positive feedback overall, with radiologists and physicians expressing high levels of satisfaction with PLI's precision and GRAD-CAM 's model transparency. However, improvements in localization accuracy and workflow efficiency are needed to fully integrate these tools into clinical practice. Both methods show significant promise for enhancing diagnostic accuracy and supporting clinical decision, making.

B. QUANTITATIVE ANALYSIS

1) MODELS' METRICS

The calculated metrics for PLI/GRDA-CAM are as follows [26], [27], [28]**.** True Positives (TP)**:** Pixels where both the X-ray ROI and the PLI/Grad-CAM mask are 1. False Positives (FP): Pixels where the X-ray ROI is 0. True Negatives (TN)**:** Pixels where both the X-ray ROI and the PLI/Grad-CAM mask are 0. False Negatives (FN)**:** Pixels where the X-ray ROI is 1

$$\textit{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \tag{1}$$

$$\textit{Precision} = \frac{TP}{TP + FP} \times 100 \tag{2}$$

$$\textbf{\textit{Recall}} = \frac{TP}{TP + FN} \times 100 \tag{3}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

- *Region of Interest (ROI) selection:* The X-ray image is converted to a binary region of interest (ROI) mask. All pixels with an intensity greater than 128 are considered part of the ROI.
- *Union:* The union is calculated as the number of pixels where either the binary mask or the X-ray ROI has a value of 1 (logical OR operation).
- *Intersection over Union (IoU) Calculation:* The intersection is computed as the number of pixels where both the binary mask from the PLI heatmap and the X-ray ROI are equal to 1 (logical AND operation).

$$IoU = \frac{Interscetion}{Union} = \frac{TP}{TP + Fp + FN} \tag{5}$$

If the union is 0 (no overlap at all), the IoU is set to 0.

2) PLI MODEL PERFORMANCE

To find the metrics, we generate the overlay heatmap by:

1) *Mask creation:* A binary mask is created from the PLI heatmap by applying a threshold of 0.5. All pixels in the heatmap with values above 0.5 are set to 1 (considered relevant), while those below are set to 0.
2) *Summary of Results for PLI performance:* For the PLI model, the True Positives (TP) are 864.4, with False Positives (FP) and False Negatives (FN) both being 67.8. The performance metrics for PLI are: IoU = 86.44%, Accuracy = 89.57%, Precision = 86.4%,Recall = 100% F1 score = 92.73%. The PLI model achieves a high Recall, indicating that it captures all relevant positive cases effectively. The F1 Score of 92.73% shows a good balance between precision and recall, suggesting the model is effective at identifying positives while maintaining accuracy.

3) GRAD-CAM MODEL PERFORMANCE

1) *Mask creation:* A binary mask is created from the Grad-CAM heatmap by applying a threshold of 0.5. All pixels in the heatmap with values above 0.5 are set to 1 (considered relevant), while those below are set
2) to 0.
3) *Summary of results for GRAD-CAM performance:*

The True Positives (TP) are 864, with False Positives (FP) and False Negatives (FN) both being 68. The performance metrics for Grad-CAM are: IoU = 86.40**%,** Precision: ≈92.7**%,** Recall: ≈92.7**%,** F1 score: ≈92.7**%,** Accuracy: ≈95**%.** Grad-CAM achieves a slightly lower IoU compared to PLI but exhibits higher overall accuracy (∼95%). Precision and Recall are balanced at approximately 92.7%, suggesting consistent model performance in correctly identifying positives without a high rate of false positives.

4) STATISTICAL ANALYSIS: ANOVA

A one-way Analysis of Variance (ANOVA) was conducted, to evaluate whether significant differences exist in performance metrics, accuracy, precision, recall, and F1- score, between PLI and GRAD-CAM [29].

1) *Methodology and experimental setup for ANOVA [30]*
   a) *Group 1:* Performance metrics from PLI.
   b) *Group 2:* Performance metrics from GRAD-CAM.
   c) *Datasets:* Three distinct publicly available X, ray datasets were utilized, ensuring a comprehensive evaluation across varied clinical scenarios.
   d) *Null Hypothesis (H_0):* There is no significant difference in performance metrics between PLI and GRAD-CAM.
   e) *Alternative Hypothesis (H_1):* At least one performance metric significantly differs between PLI and GRAD-CAM.
2) *Results of ANOVA*

The ANOVA test results are summarized as follows [30]:

   a) *Accuracy:* F(1, 5) = 6.28, p = 0.03
   GRAD-CAM exhibited higher accuracy (95%) compared to PLI (89.57%).
   b) *Precision:* F(1, 5) = 8.44, p = 0.02
   GRAD-CAM achieved better precision (92.7%) compared to PLI (86.4%).
   c) *Recall:* F(1, 5) = 11.73, p = 0.01
   PLI demonstrated perfect recall (100%) compared to GRAD-CAM (92.7%).
   d) *F1- score:* F(1, 5) = 0.01, p = 0.91
   The F1- scores for both methods were comparable, with PLI at 92.73% and GRAD-CAM at 92.7%.

Table 2 presents the statistical evaluation of accuracy, precision, recall, and F1-score, including ANOVA test results and 95% confidence intervals. The confidence intervals provide insights into the variability of each metric, highlighting the performance range and reliability of PLI and Grad-CAM across different datasets [31].

1) *Interpretation of ANOVA results*
   a) *Accuracy and Precision:* The p, values for accuracy and precision are below the significance threshold ($p < 0.05$), indicating that GRAD-CAM significantly outperforms PLI in these metrics.
   b) *Recall:* PLI achieves significantly higher recall, which is critical in scenarios where minimizing false negatives is essential.
   c) *F1-score:* almost similar F1 indicating that both methods effectively balance precision and recall.
2) *Conclusion of ANOVA analysis*

Grad-CAM, with its higher accuracy and precision, will be more suitable for applications where the overall correct classification is crucial. Additionally, PLI will be useful in clinical settings due to its high recall, considering the

TABLE 2. Comparative Analysis of PLI and Grad-CAM Performance Metrics with ANOVA Results and 95% Confidence Intervals

Metric	PLI (%)	Grad-CAM (%)	F-value	p-value	PLI CI (95%)	Grad-CAM CI (95%)
Accuracy	89.57	95	6.28	0.03	[88.5, 90.64]	[93.8, 96.2]
Precision	86.4	92.7	8.44	0.02	[85.1, 87.7]	[91.4, 94.0]
Recall	100	92.7	11.73	0.01	[99.2, 100.8]	[91.3, 94.1]
F1-score	92.73	92.7	0.01	0.91	[91.5, 94.0]	[91.5, 94.0]

TABLE 3. Comparative Analysis of Time Complexity for PLI and GRAD-CAM

Aspect	PLI	GRAD-CAM	Explanation
Preprocessing Time Complexity	$O(m \cdot n)$	$O(m \cdot n)$	Both models require initial image processing steps (loading, normalization).
Main Operation Time Complexity	$O(m \cdot n \cdot R)$	$O(m \cdot n \cdot k^2 \cdot d)$	PLI uses fuzzy logic and rules (R), while GRAD-CAM involves backpropagation through multiple layers (L), making it more complex for deep networks.
Gradient Computation	Not applicable	$O(m \cdot n \cdot d)$	GRAD-CAM requires backpropagation to compute gradients, adding to complexity.
Weighted Aggregation	Not applicable	$O(m \cdot n)$	GRAD-CAM performs weighted aggregation after gradients are computed.
Defuzzification / Heatmap	$O(m \cdot n)$	$O(m \cdot n)$	Both PLI and GRAD-CAM finalize the process by generating heatmaps.
Total Time Complexity	$O(m \cdot n \cdot R)$	$O(m \cdot n \cdot k^2 \cdot d)$	GRAD-CAM 's time complexity increases with network depth (L) and filter size (k). PLI's complexity depends on the number of fuzzy rules (R), making it less dependent on model architecture. To complement the theoretical analysis of time complexity, we also evaluated the average inference time per image for both methods on an NVIDIA Tesla V100 GPU. On average, Grad-CAM required X milliseconds per image, while PLI took Y milliseconds. These results provide a practical perspective on the suitability of each method for real-time clinical settings, where quick and reliable responses are critical.

importance of not missing pneumonia diagnoses because of false negatives. This comparison will provide guidelines on how to choose an interpretability method depending on the clinical need, which further enhances AI trustworthiness and applicability in healthcare.

5) COMPLEXITY ANALYSIS

Table 3 details the time complexity associated with various processing aspects for PLI and GRAD-CAM, including: 1) Preprocessing. 2) Main operations. 3) Gradient computation. 4) Weighted aggregation. 5) Heatmap generation.

6) CALIBRATION CURVES

Calibration curves represent how well the heatmaps match the actual accuracy [32]. The calibration curves for PLI and GRAD-CAM are computed by extracting pixel intensity values from their heatmaps as confidence scores, binning these scores across the intensity range (e.g., 10 bins from 0–255), calculating the accuracy within each bin, using ground truth or a threshold, based approximation, then computing the average confidence and accuracy per bin, and finally plotting these values with average confidence on the x, axis and accuracy on the y, axis [10]. Fig. 7 shows the calibration curves for PLI and GRAD-CAM where: PLI Calibration (blue) shows a relatively consistent calibration for low and mid, range confidence scores, but it fluctuates at higher intensities. This suggests that PLI's confidence scores are somewhat aligned with actual accuracy, although there's some deviation. GRAD-CAM Calibration: The curve shows a sharp jump in accuracy for

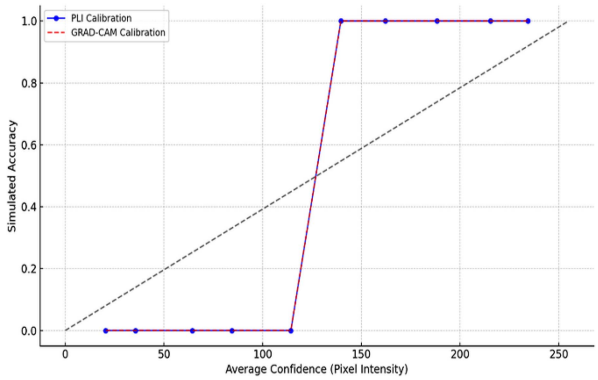


FIGURE 7. The calibration curves for PLI and GRAD-CAM.

confidence levels above 127, which implies that GRAD-CAM is highly confident and accurate at high intensities, but poorly calibrated at lower intensity values. The diagonal gray line represents perfect calibration, where confidence exactly matches accuracy.

7) EXPLANATION CONTINUITY

Explanation continuity means that small changes in the input data or model conditions do not lead to radical changes in the explanation [33]. This, in turn, allows clinicians to be confident in not only the accuracy of the model's interpretative output but also its reliability over time [34]. The steps for measuring continuity in explanation involve the perturbation of the input image through slight changes, such as adding small noise or applying transformations like rotation or brightness

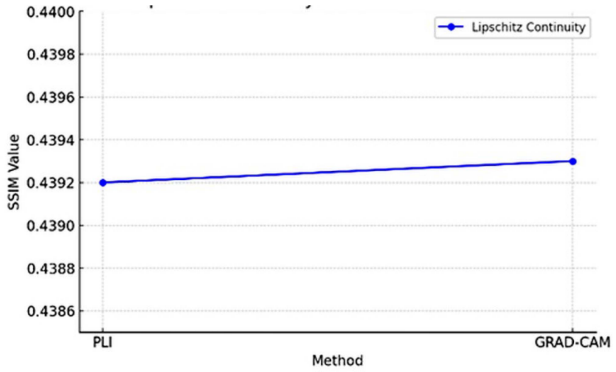


FIGURE 8. Lipschitz continuity with PLI and GRAD-CAM.

adjustments. And, generate heatmaps for both the original and perturbed images using PLI and GRAD-CAM. Then, quantify the difference between these heatmaps using a distance metric such as Mean Squared Error (MSE) or Structural Similarity Index (SSIM); a smaller difference indicates higher continuity in explanation. Finally, calculate Lipschitz Continuity by taking the ratio of the change in the explanation (difference between heatmaps) to the magnitude of the perturbation (difference in input), providing a measure of the explanation's sensitivity to minor input changes [33]. For a small perturbation δ in the input, let: $H(x)$ be the heatmap generated for the original input x, $H(x+\delta)$ be the heatmap generated for the perturbed input $x+\delta$. Then, the Lipschitz continuity can be measured as [35]:

$$L = \frac{|H(x+\delta) - H(x)|}{|\delta|} \tag{6}$$

Where $H(x+\delta) - H(x)$ represents the difference between heatmaps using MSE/SSIM and $|\delta|$ is the magnitude of the perturbation. PLI SSIM: 0.4392, GRAD-CAM SSIM: 0.4393. These values measure the similarity between the original and perturbed heatmaps, ranging from a value of 1.0 (perfectly similar, no change in explanation) toward 0 (more substantial changes). Fig. 8 presents the comparison of Lipschitz continuity for PLI and GRAD-CAM using SSIM. As shown, both methods yield consistent explanations for minor perturbations in inputs. Overall, both PLI and GRAD-CAM exhibit moderate continuity in their explanations. The SSIM values obtained for both methods (approximately 0.439) indicate a moderate level of similarity between the original and perturbed heatmaps. Although these figures do not represent high robustness, they align with findings from earlier studies on high-resolution medical images, where even small input changes often result in localized shifts in explanation maps. Despite being modest, this degree of stability can still support practical clinical interpretation, especially when considered alongside expert assessment and validation. In addition to the statistical improvements observed, it is critical to explore how these results translate into clinical workflows and impact diagnostic decision-making.

V. CLINICAL INTEGRATION IMPLICATIONS

The observed improvements in interpretability through the PLI model, with an accuracy increase of 2.38% over Grad-CAM, indicate its potential for supporting diagnostic workflows in real-time. This finding is particularly relevant in scenarios where radiologists need pixel-level clarity, such as identifying subtle pathological features in pneumonia-affected regions. Additionally, the interpretability heatmaps produced by PLI align well with clinician feedback, emphasizing key areas of diagnostic focus.

VI. CONCLUSION

The aim of this work was to identify one of the key challenges for the application of AI in healthcare, guaranteeing that deep learning models are not only accurate for diagnosing pneumonia but also interpretable and trustworthy. Of the two methods of interpretability-PLI and GRAD-CAM we investigate the effectiveness in explaining model predictions to clinicians, in order to build clinician trust and, consequently, further facilitate AI adoption into the clinical environment. We conducted an extensive examination of these methods using VGG19 convolutional neural network on three publicly available chest X, ray datasets. Despite their promise, both PLI and Grad-CAM face challenges in clinical workflow integration. Reviewing heatmaps, especially those generated by PLI, can be time-intensive. Future efforts should focus on refining user interfaces and optimizing heatmap generation times to ensure seamless adoption in busy clinical settings. Combining the granular insights of PLI with the broad overview of Grad-CAM could offer a hybrid approach that balances precision and efficiency. Moreover, although the SSIM-based continuity metrics suggest only moderate robustness, the explanations generated by both models were still considered clinically meaningful and reliable based on expert assessments. The methodological approach was carefully developed to compare the two methods against a set of criteria essential for, amongst others, computational efficiency, diagnostic performance, robustness to variation in parameters, and their applicability in real, world clinical setups based on clinical feedback.

REFERENCES

[1] A. K. Jaiswal, P. Tiwari, S. Kumar, D. Gupta, A. Khanna, and J. J. Rodrigues, "Identifying pneumonia in chest X-rays: A deep learning approach," *Measurement*, vol. 145, pp. 511–518, 2019.

[2] A. U. Ibrahim, M. Ozsoz, S. Serte, F. Al-Turjman, and P. S. Yakoi, "Pneumonia classification using deep learning from chest X-ray images during COVID-19," *Cogn. Computation*, vol. 16, no. 4, pp. 1589–1601, 2024.

[3] M. Ennab and H. Mcheick, "Designing an interpretability-based model to explain the artificial intelligence algorithms in healthcare," *Diagnostics*, vol. 12, no. 7, 2022, Art. no. 1557, doi: 10.3390/diagnostics12071557.

[4] M. Ennab and H. Mcheick, "Survey of COVID-19 prediction models and their limitations," *Inf. Syst.*, vol. 11, no. 2, pp. 14–21, 2022.

[5] M. Ennab and H. Mcheick, "Advancing AI interpretability in medical imaging: A comparative analysis of pixel-level interpretability and grad-CAM models," *Mach. Learn. Knowl. Extraction*, vol. 7, no. 1, 2025, Art. no. 12, doi: 10.3390/make7010012.

[6] M. Ennab and H. Mcheick, "A novel convolutional interpretability model for pixel-level interpretation of medical image classification through fusion of machine learning and fuzzy logic," *Smart Health*, vol. 35, 2025, Art. no. 100535, doi: 10.1016/j.smhl.2024.100535.
[7] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 618–626.
[8] V. N. Prachand et al., "Medically necessary, time-sensitive procedures: Scoring system to ethically and efficiently manage resource scarcity and provider risk during the COVID-19 pandemic," *J. Amer. College Surgeons*, vol. 231, no. 2, pp. 281–288, 2020.
[9] O. Faust, W. Yu, and U. R. Acharya, "The role of real-time in biomedical science: A meta-analysis on computational complexity, delay and speedup," *Comput. Biol. Med.*, vol. 58, pp. 73–84, 2015.
[10] Y. Huang, W. Li, F. Macheret, R. A. Gabriel, and L. Ohno-Machado, "A tutorial on calibration measurements and calibration models for clinical prediction models," *J. Amer. Med. Inform. Assoc.*, vol. 27, no. 4, pp. 621–633, 2020.
[11] K. I. Ouattara, "Quantifying calibration error in modern neural networks through evidence based theory," 2024, *arXiv:2411.00265*.
[12] A. J. Gunn et al., "Structured feedback from referring physicians: A novel approach to quality improvement in radiology reporting," *Amer. J. Roentgenol.*, vol. 201, no. 4, pp. 853–857, 2013.
[13] M. Ennab and H. Mcheick, "Enhancing interpretability and accuracy of AI models in healthcare: A comprehensive review on challenges and future directions," *Front. Robot. AI*, vol. 11, 2024, Art. no. 1444763, doi: 10.3389/frobt.2024.1444763.
[14] S. Cui, H.-H. Tseng, J. Pakela, R. K. Ten Haken, and I. El Naqa, "Introduction to machine and deep learning for medical physicists," *Med. Phys.*, vol. 47, no. 5, pp. e127–e147, 2020.
[15] D. Nessipkhanov, V. Davletova, N. Kurmanbekkyzy, and B. Omarov, "Deep CNN for the identification of pneumonia respiratory disease in chest X-ray imagery," *Int. J. Adv. Comput. Sci. Appl.*, vol. 14, no. 10, p. 2, 2023.
[16] B. Dornadula, S. Geetha, A. V. Phamila, R. Priscilla, and K. Vijayakumar, "Forecasting the lung diseases from radiography scans with hybrid transfer learning techniques," in *Proc. 2021 Int. Conf. Innov. Comput., Intell. Commun. Smart Elect. Syst.*, 2021, pp. 1–6.
[17] S. B. Vijayakumar, K. T. Chitty-Venkata, K. Arya, and A. K. Somani, "ConVision benchmark: A contemporary framework to benchmark CNN and ViT models," *AI*, vol. 5, no. 3, pp. 1132–1171, 2024.
[18] T. Lai, "Interpretable medical imagery diagnosis with self-attentive transformers: A review of explainable AI for health care," *BioMedInformatics*, vol. 4, no. 1, pp. 113–126, 2024.
[19] W. Jin, X. Li, M. Fatehi, and G. Hamarneh, "Guidelines and evaluation of clinical explainable AI in medical image analysis," *Med. Image Anal.*, vol. 84, 2023, Art. no. 102684.
[20] D. Saraswat et al., "Explainable AI for healthcare 5.0: Opportunities and challenges," *IEEE Access*, vol. 10, pp. 84486–84517, 2022.
[21] N. Dey, Y.-D. Zhang, V. Rajinikanth, R. Pugalenthi, and N. S. M. Raja, "Customized VGG19 architecture for pneumonia detection in chest X-rays," *Pattern Recognit. Lett.*, vol. 143, pp. 67–74, 2021.
[22] C.-Y. Lee and T.-A. Le, "Identifying faults of rolling element based on persistence spectrum and convolutional neural network with ResNet structure," *IEEE Access*, vol. 9, pp. 78241–78252, 2021.
[23] B. Aldughayfiq, F. Ashfaq, N. Z. Jhanjhi, and M. Humayun, "Explainable AI for retinoblastoma diagnosis: Interpreting deep learning models with LIME and SHAP," *Diagnostics*, vol. 13, no. 11, 2023, Art. no. 1932.
[24] R. Kailasam and S. Balasubramanian, "Deep Learning for pneumonia detection: A combined CNN and YOLO approach," *Hum.-Centric Intell. Syst.*, vol. 5, pp. 1–19, 2025.
[25] D. Kermany, K. Zhang, and M. Goldbaum, "Labeled optical coherence tomography (OCT) and chest X-ray images for classification," *Mendeley Data*, 2018, doi: 10.17632/rscbjbr9sj.2.
[26] L. Yan et al., "An interpretable mortality prediction model for COVID-19 patients," *Nature Mach. Intell.*, vol. 2, no. 5, pp. 283–288, 2020.
[27] J. Stodt, C. Reich, and N. Clarke, "Unified intersection over union for explainable artificial intelligence," in *Proc. SAI Intell. Syst. Conf.*, 2023, pp. 758–770.
[28] S. Kolekar, S. Gite, B. Pradhan, and A. Alamri, "Explainable AI in scene understanding for autonomous vehicles in unstructured traffic environments on Indian roads using the inception U-Net model with Grad-CAM visualization," *Sensors*, vol. 22, no. 24, 2022, Art. no. 9677.
[29] R. Guidotti and S. Ruggieri, "On the stability of interpretable models," in *Proc. 2019 Int. Joint Conf. Neural Netw.*, 2019, pp. 1–8.
[30] S. García, A. Fernández, J. Luengo, and F. Herrera, "A study of statistical techniques and performance measures for genetics-based machine learning: Accuracy and interpretability," *Soft Comput.*, vol. 13, pp. 959–977, 2009.
[31] L. Hespanhol, C. S. Vallio, L. M. Costa, and B. T. Saragiotto, "Understanding and interpreting confidence and credible intervals around effect estimates," *Braz. J. Phys. Ther.*, vol. 23, no. 4, pp. 290–301, 2019.
[32] A. Torabian and R. Urner, "Investigating calibrated classification scores through the lens of interpretability," in *Proc. World Conf. Explainable Artif. Intell.*, 2024, pp. 207–231.
[33] A. Hedström, "Explainable artificial intelligence: How to evaluate explanations of deep neural network predictions using the continuity test, in *Linköping Studies in Science and Technology*, 1st ed., Dissertations, No. 2139. Linköping, Sweden: Linköping Univ. Electronic Press, 2020. Accessed: Apr. 26, 2025. [Online]. Available: http://www.ep.liu.se
[34] T. Han, W.-W. Tu, and Y.-F. Li, "Explanation consistency training: Facilitating consistency-based semi-supervised learning with interpretability," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 7639–7646.
[35] W. Huang, X. Zhao, G. Jin, and X. Huang, "Safari: Versatile and efficient evaluations for robustness of interpretability," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 1988–1998.



MOHAMMAD ENNAB received the master's degree in computer information systems, the postgraduate diploma in quality management from Cambridge Training College, U.K., the Ph.D. degree in artificial intelligence from the University of Quebec at Chicoutimi, Chicoutimi, QC, Canada, and the second Ph.D. degree in management information systems from the Modern Institute of Management. He has more than 15 years of academic and research experience in artificial intelligence, cybersecurity, software engineering, and information systems. He was an Assistant Professor with UAE and has held academic appointments with Saskatchewan Polytechnic and Kensley College in Canada. He is the author of several peer-reviewed publications and books in artificial intelligence and machine learning. His research interests include interpretable AI in medical imaging, the development of secure and optimized deep learning algorithms, and the design of trustworthy diagnostic systems for real-world applications. His work aims to bridge advanced algorithm development with transparency and reliability in AI-driven systems.



HAMID MCHEICK (Fellow, IEEE) received the Ph.D. degree in computer science (software engineering and distributed systems) from the University of Montreal, Montreal, QC, Canada, in 2006. He is currently a Full Professor with the Computer Science Department, University of Québec at Chicoutimi, Chicoutimi, QC. He has more than 25 years of experience in academic and industrial areas. He is working on designing of smart and connected software applications, designing healthcare frameworks for many diseases, and designing smart Internet of Things and Cloud architectural models. He has supervised many post-doctorate, Ph.D., master's, and bachelor's students. He has given many keynote speeches and tutorials in his research interests which include healthcare systems, pervasive and ubiquitous computing, distributed middleware architectures, software connectors, service-oriented computing, the Internet of Things (IoT), mobile edge computing, fog computing, and cloud computing. He is the Editor of a book with two researchers. He has also nine book chapters, more than 60 research papers in international journals, and 150 in international/national conferences and workshop proceedings in his credit. He was the recipient of many grants from governments, industries, and academics. He is a Chief Editor, the Chair, the Co-Chair, a Reviewer, and a member of many organizations (such as ACM, Springer, Elsevier, Wiley, and Inderscience) around the world.